

Privacy by Design

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of data retention and deletion.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 28: Privacy by Design

Data minimization, retention rules, and consent belong in the architecture.



CREATOR MISSION

Remove unnecessary student data from the design.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Sandboxing and resource limits. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can explain privacy by design using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use privacy by design to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

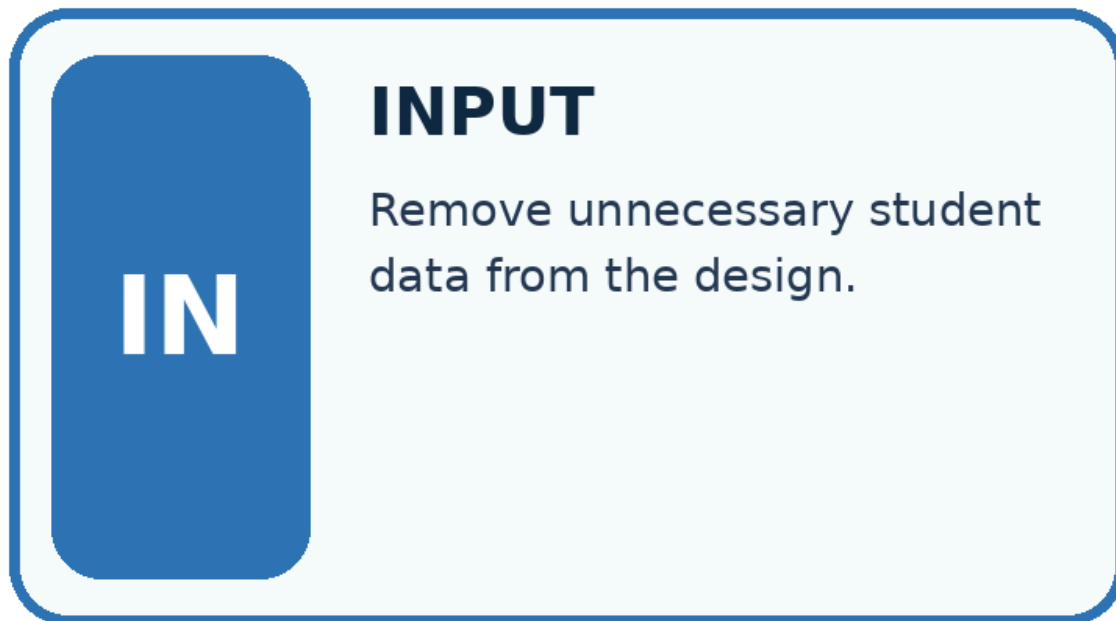
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Data minimization, retention rules, and consent belong in the architecture.





EVIDENCE

Remove unnecessary student data from the design.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Data retention and deletion — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Record ages are 6, 7, 9 days. Cache contains copies of all three.

Check or calculation: Toy policy retains session content for 7 days and deletes it when age is at least 7.

Conclusion and limit: Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: The cache deletion fails.

Prediction: Report incomplete deletion rather than success.

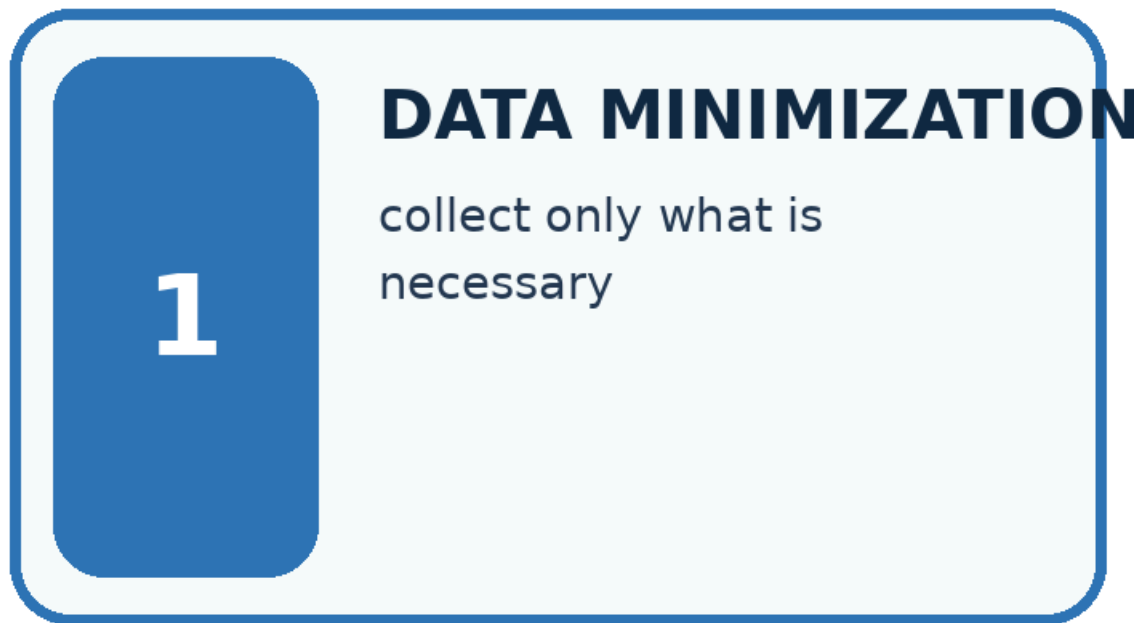
Reason: Removing the working copy does not remove the known retained cache copy.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

CONSENT

informed permission for a defined use

3

RETENTION

how long data is kept

4

ACCESS CONTROL

who can view or change
data

Word · My own example · Not this / boundary

Word: data minimization

My own example: Collecting and retaining only the information needed for a specified purpose.

Not this / boundary: _____

Word: consent

My own example: An informed and freely given agreement to a specific use or action.

Not this / boundary: _____

Word: retention

My own example: Keeping information for a period of time rather than deleting it immediately.

Not this / boundary: _____

Word: access control

My own example: Rules and mechanisms deciding who may view information or perform an action.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Retention period

Meaning: The declared lifetime of stored information

Word: Deletion boundary

Meaning: The age at which removal applies

Word: Residual copy

Meaning: A remaining duplicate in another store

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 28 • WORKED EVIDENCE

- 1 Study help needs topic and current question—not home address.
- 2 Keep session state temporarily; do not store by default.
- 3 Let users view and delete saved preferences.
- 4 Protect logs from containing private source text or identifiers.

1. Study help needs topic and current question—not home address.

2. Keep session state temporarily; do not store by default.

3. Let users view and delete saved preferences.

4. Protect logs from containing private source text or identifiers.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Record ages are 6, 7, 9 days. Cache contains copies of all three.

Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Privacy by Design

Trace deletion through both a working folder and a cache.

The facts we will use

Fictional retention policy: delete records aged at least 3 days. Ages A=2, B=3, C=4; the cache contains a copy of B.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Privacy by Design

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare ages

A is below 3; B is exactly 3; C is above 3.



2. Delete eligible

Keep A. Delete B and C from the working folder.



3. Check copies

Delete cached B too; otherwise the declared content remains retained.

CHANGE ONE THING

The cache deletion fails.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compare ages

What happens and why: A is below 3; B is exactly 3; C is above 3.

Step: 2. Delete eligible

What happens and why: Keep A. Delete B and C from the working folder.

Step: 3. Check copies

What happens and why: Delete cached B too; otherwise the declared content remains retained.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

This fictional policy is a tracing exercise, not a statement of legal retention requirements.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Privacy can be added after the product works. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

This fictional retention exercise is not a statement of legal retention requirements.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Data retention and deletion”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Data retention and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which IDs and locations must appear in the deletion evidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Audit every field for purpose, permission, retention, access, and deletion; remove or shorten anything not essential.

Audit every field for purpose, permission, retention, access, and deletion; remove or shorten anything not essential.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Create a minimal data map and visible memory controls before expanding features.

THIS WEEK’S CHECKPOINT Create a minimal data map and visible memory controls before expanding features.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Remove unnecessary student data from the design.

CHALLENGE Remove unnecessary student data from the design.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A study tool plans to retain names, addresses and all chats forever.

The question

Design a smaller data plan.

Worked explanation

Keep only needed preferences or progress, with retention, deletion and access controls. Avoid unnecessary addresses and raw chat storage.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Keep a small, purposeful data plan

Evidence or stage · Value or result

Evidence or stage: Preference

Value or result: Reading level may be useful

Evidence or stage: Address

Value or result: Unnecessary for chapter study

Evidence or stage: Raw chats

Value or result: Avoid indefinite retention

Evidence or stage: Controls

Value or result: Define access, retention and deletion

A study tool plans to retain names, addresses and all chats forever.

Design a smaller data plan.

Keep only needed preferences or progress, with retention, deletion and access controls. Avoid unnecessary addresses and raw chat storage.

14. Try a new case independently

Independent case: Apply a new case independently

Retention is 5 days, inclusive deletion boundary. Ages 4, 5, 6; a log stores complete content. Trace required action.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A practice quiz needs a topic and difficulty. A form also asks for home address and date of birth. What should be removed and why?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create a minimal data map and visible memory controls before expanding features.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A study tool needs topic and reading level but proposes storing addresses and all conversations forever. Design a smaller record and a deletion rule.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.